Total No. of Questions : 8] [Total No. of Printed Pages : 4

Paper Code : 21317

F-417

B.C.A. (Fourth Semester) Examination, 2023

(New Course)

Paper No. BCA-N-402

INTRODUCTION TO DBMS AND SQL

Time : Three Hours] *[Maximum Marks : 70*

Note : Attempt any **five** questions. **All** questions carry equal marks.

1. (a) Discuss three level of abstraction or schema architecture of DBMS? 7

   (b) What is the role of database administrator in database management system? Describe weak and strong entity set in database with examples? 7

2. (a) Explain all database languages in detail with examples? 7

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   (b) Explain Data modeling in database. Why would you choose a database system instead of simply storing data in operating system files? 7

3. (a) Explain the Primary Key, Super Key, Foreign Key and Candidate Key with examples? 7

   (b) What is Normalization? How 4NF is differing from PJNF? Explain with examples? 7

4. (a) Consider the following relations : 7

   Hotel {hotelNo, name, address} Room {roomNo, hotelNo, type, price}

   Booking {hotelNo, guestNo, dateFrom, dateTo, roomNo}

   Guest {guestNo, name, address}

   Write the SQL statements for the following:

   (i) List the names and addresses of all guests in Chandigarh, alphabetically ordered by name.

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(ii) List all family rooms with a price below Rs. 400 per night, in ascending order of price.

(iii) How many Hotels are there.

(iv) List of Guest name with Room No in ascending order

(b) Differentiate between Hierarchical model and Network Model? 7

5. (a) How does Boyce-Codd normal form is differ from 3 NF? Why is it considered stronger than 3 NF? 7

(b) What is functional dependency? Discuss its types. Explain the role of functional dependency in the process of normalization. 7

6. (a) What are the arithmetical operators in SQL? Explain with examples? 7

(b) What do you mean by integrity constraints? Explain with examples? 7

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7. (a) Explain Normalization? What are the difference between 4NF and 5NF? 7

(b) What are the different types of anomalies associated with database? Describe with examples? 7

8. Write short notes with suitable examples : 14

(i) RDBMS

(ii) 1NF and 2NF

(iii) Transaction

(iv) Distributed database

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